ETH zürich

Mobile Computing

Emerging Technologies
Non-Terrestrial Satellite Networks

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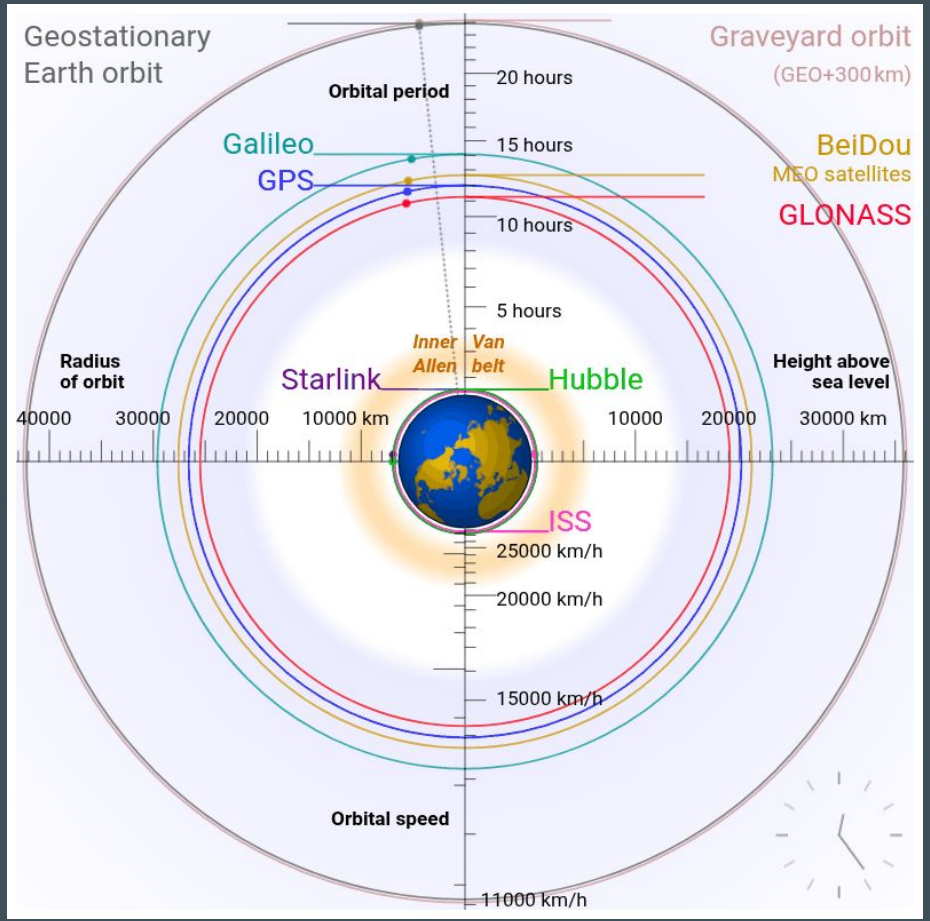
Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

Satellite Orbits



Low Earth Orbit

- 200–2,000 km above Earth
- Satellite Internet
  - Starlink, OneWeb, and similar for low-latency coverage
- Earth Observation
  - Imaging (e.g., weather satellites, climate monitoring, and remote sensing for agriculture or natural disaster mgmt.)
- Science and Research
  - Platforms like the International Space Station (ISS), Tiangong space station, Hubble

Medium Earth Orbit

- 2,000–35,786 km above Earth
- Global Navigation Systems
  - GPS, Galileo, and GLONASS satellites operate in MEO to provide precise positioning and navigation.
- Communication Satellites
  - Some systems (e.g., O3b Networks) use MEO to provide decent latency internet for underserved areas. This is becoming less relevant with the new LEO constellations.

Geostationary Earth Orbit

- 35,786 km above Earth
  - Telecommunications: TV broadcasting, satellite phones, and long-distance communication.
  - Weather Monitoring: Geostationary weather satellites like GOES continuously monitor large regions of Earth.
  - Military Communications: Used for secure and wide-area military networks.
- One GEO satellite has a theoretical line-of-sight coverage of ~42% of Earth's surface
  - Practical footprint is smaller, often described as one-third of Earth

Increasing number of payloads

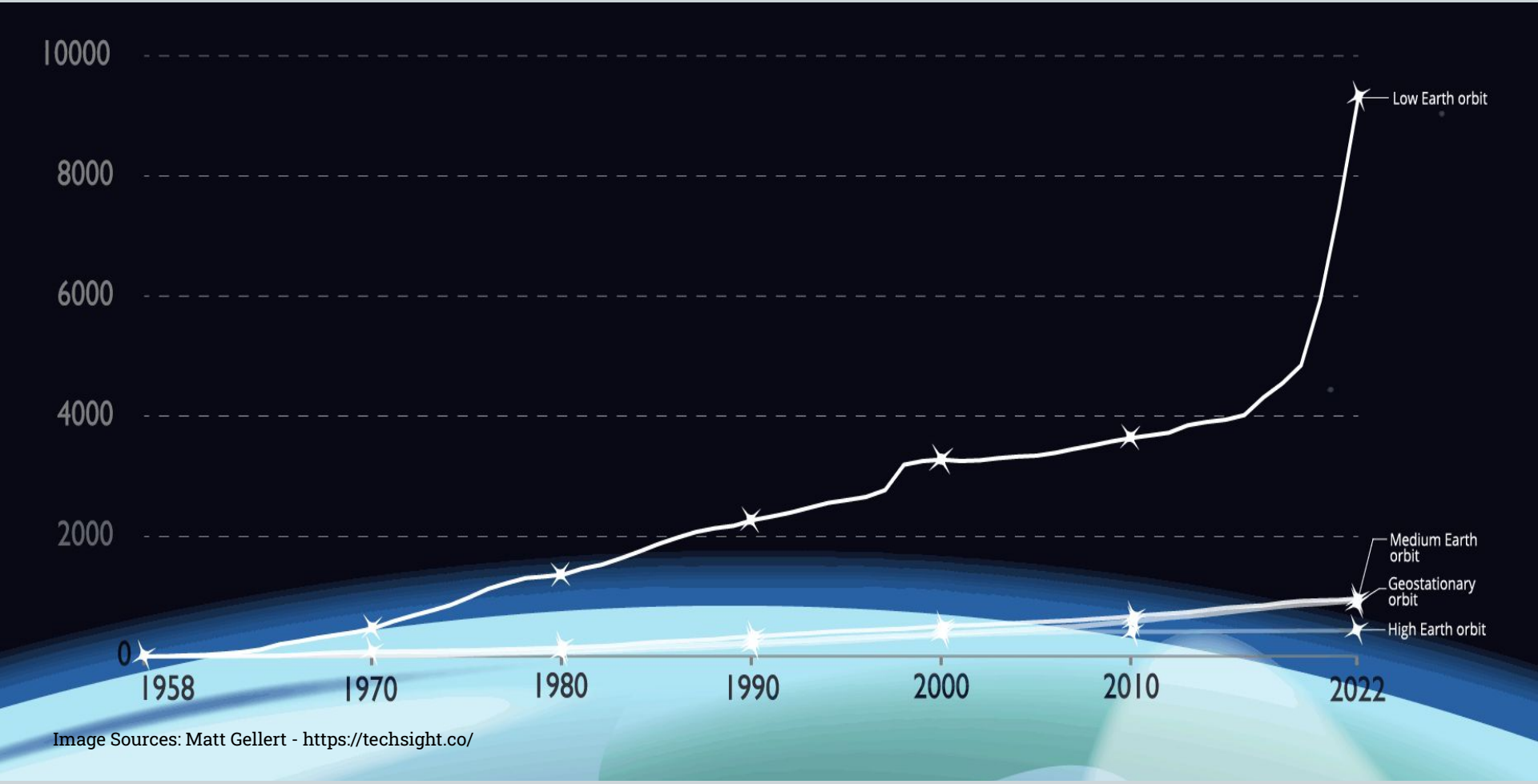


Image Sources: Matt Gellert - https://techsight.co/

Advantages of Satellite Networking

- Covers large areas, including difficult to reach locations
- Resilience to disasters compared to terrestrial infrastructure
- Speed of light ~33% faster vs fiber
  - NY-> London 22ms << 33ms

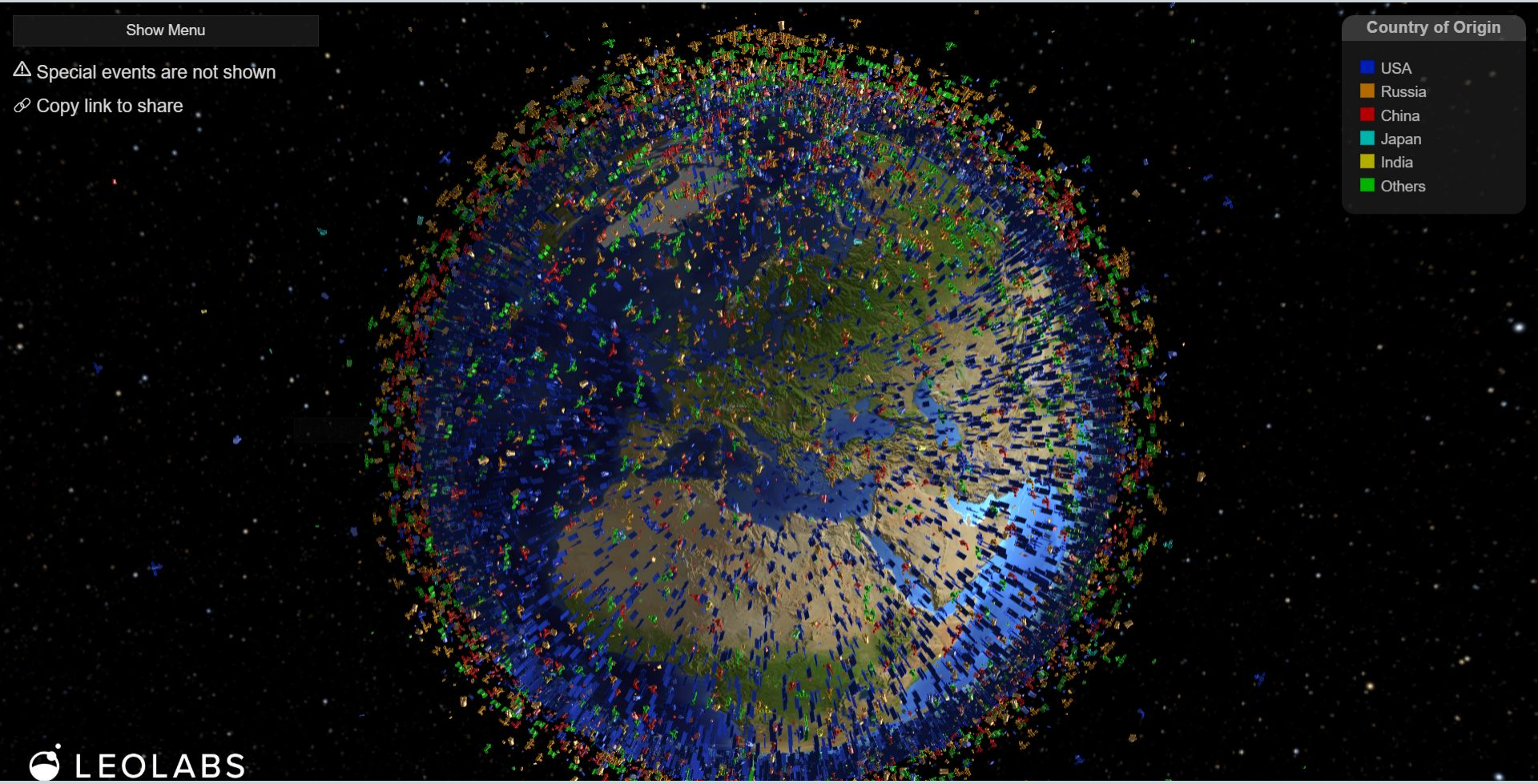
Disadvantages

- High latency (mainly for GEO)
- Cost of deployment, and maintenance
- Space debris, see for example LeoLabs Visualization
- Rain / atmospheric attenuation
- Atmospheric drag (LEO)

LEO Country of Origin: CH registered 13



All LEO Satellites: ~17000



Satellite Networking Frequencies

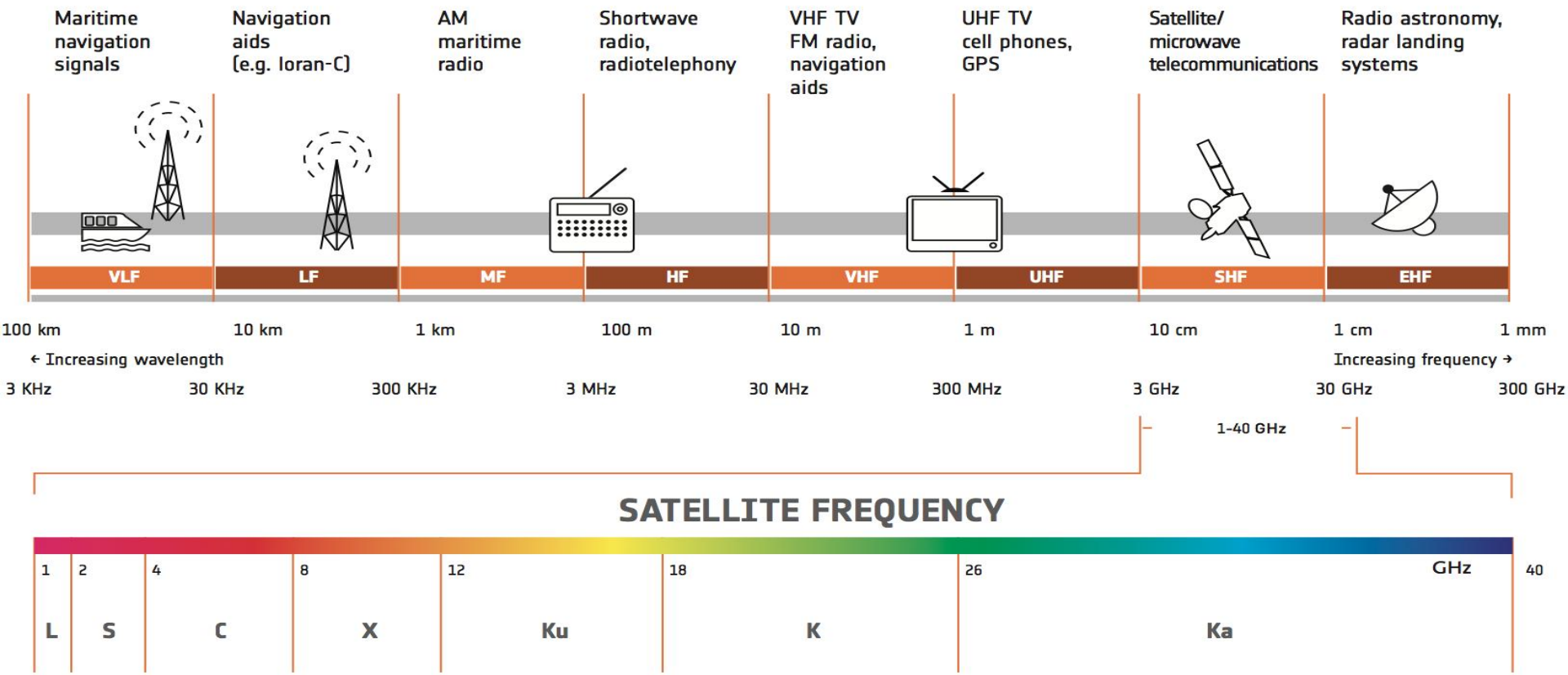
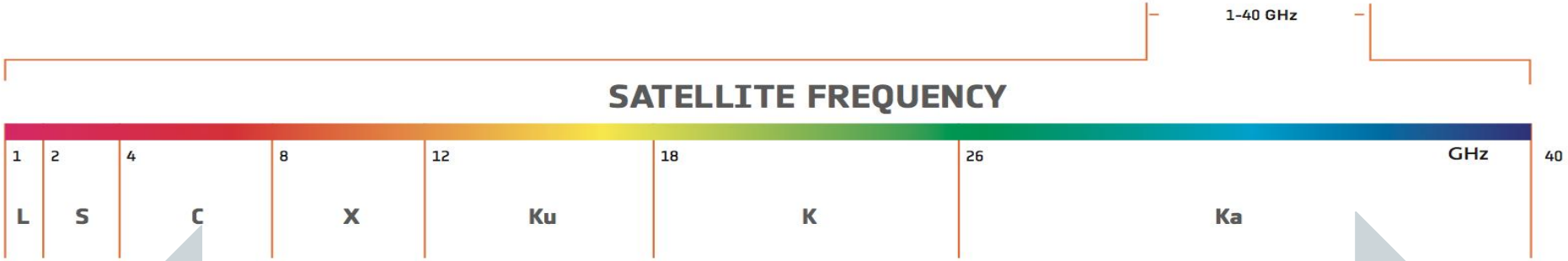


Image Sources: ESA - https://www.esa.int/Applications/Connectivity_and_Secure_Communications/Satellite_frequency_bands

Satellite Networking Frequencies



Better Propagation
Longer Range
More Reliable in Bad Weather

Higher Bandwidth
Smaller Antennas
Better for Point-to-Point
Communications

Lower Bandwidth
Larger Antennas

Higher Power Requirements
Susceptible to Rain Fade

Image Sources: ESA - https://www.esa.int/Applications/Connectivity_and_Secure_Communications/Satellite_frequency_bands

Satellite Networking Frequencies

- Different satellites can have very different requirements and applications
- GPS satellites need to send very little data in a broadcast-only manner
  - need to be resilient to atmospheric events
- TV satellites have higher bandwidth requirements
  - Require reliable, continuous coverage to a fixed area

Satellite Networking Frequencies

Satellite Type	Frequency Bands	Typical Bandwidth	Bandwidth (Mbps/Gbps)	Latency
GPS Satellites	**C-band** (3.7-4.2 GHz)	None (unused for GPS)	None	None
TV Satellites (GEO)	**C-band** (3.7-4.2 GHz)	36 MHz to 72 MHz per transponder	**~ 36-72 Mbps** (per transponder)	500 ms (round trip)
	Ku-band (12-18 GHz)	36 MHz to 72 MHz per transponder	**~ 72 Mbps to 500 Mbps** (per transponder)	500 ms (round trip)
	Ka-band (26.5-40 GHz)	500 MHz to 1 GHz per transponder	**~ 500 Mbps to 1 Gbps** (per transponder)	500 ms (round trip)

Ground Stations



NASA Deep Space Station 35, Australia



Image Sources: Canberra Deep Space Communication Complex - https://www.csiro.au/



S-band uplink: 16 kb/s
Commanding

S-band downlink: 40 kb/s
Ranging

Ka-band downlink: 28 Mb/s
Stored science and engineering data
Telemetry

DEEP SPACE NETWORK (DSN)



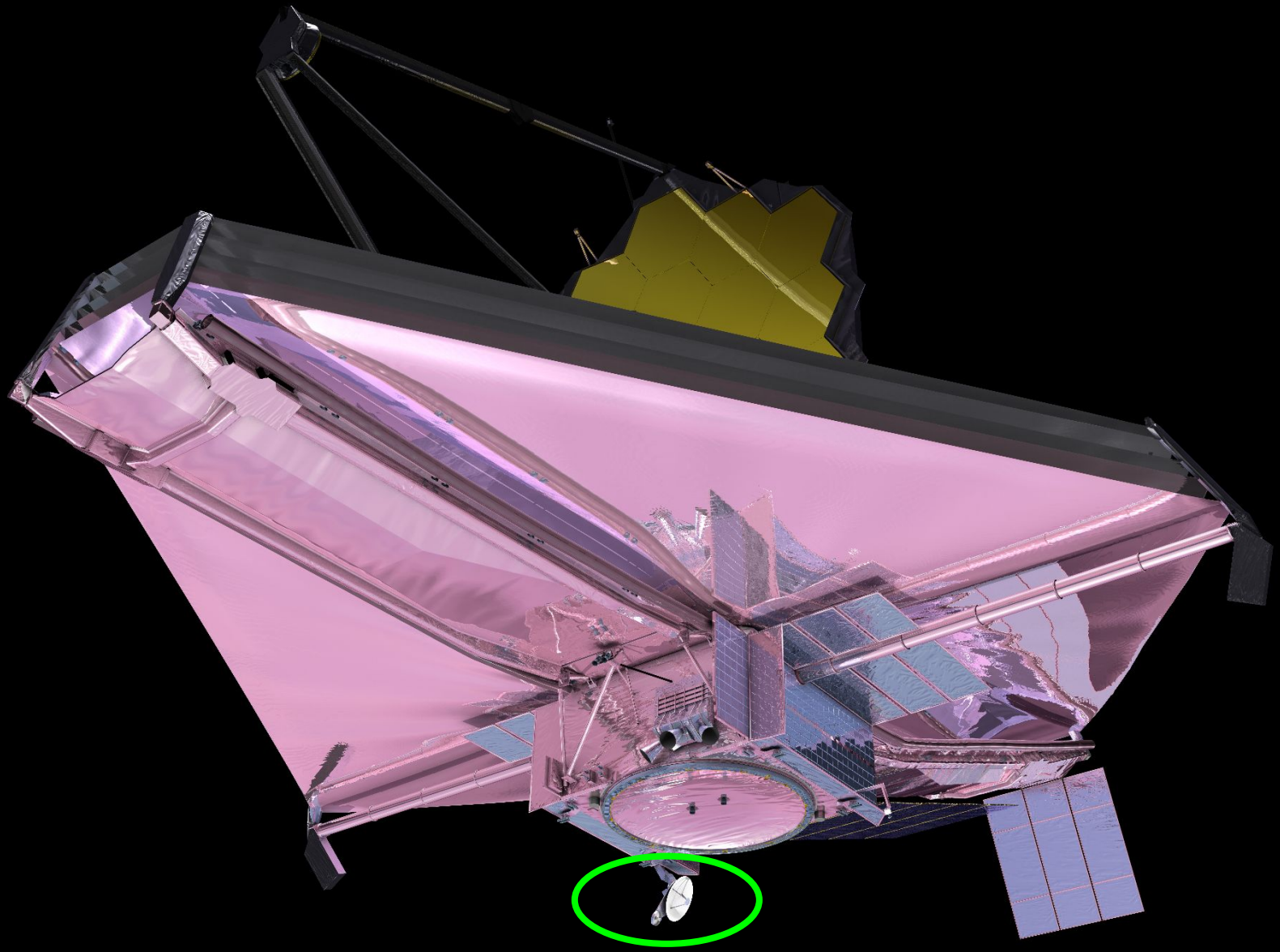
Goldstone, CA, USA

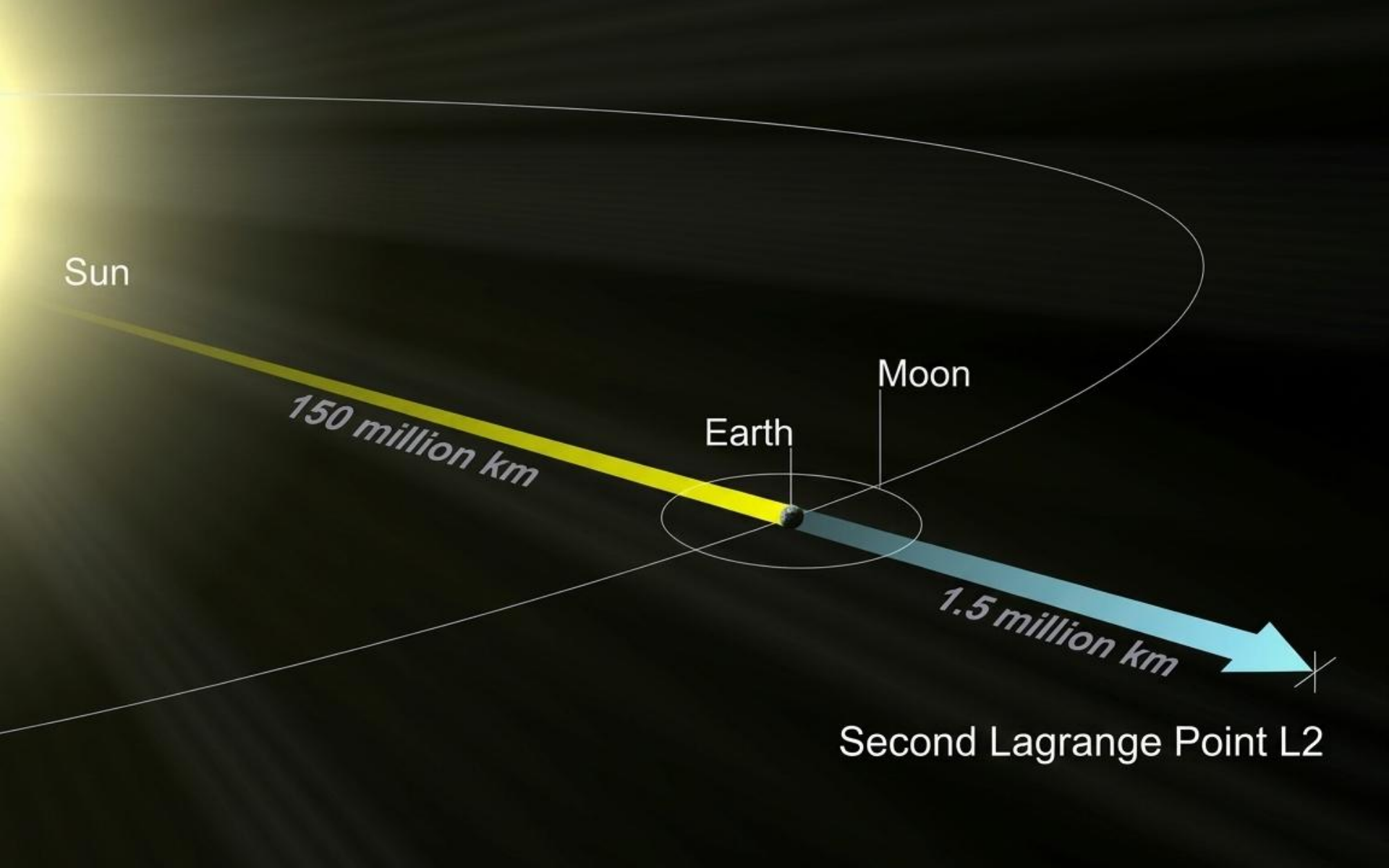


Canberra, Australia

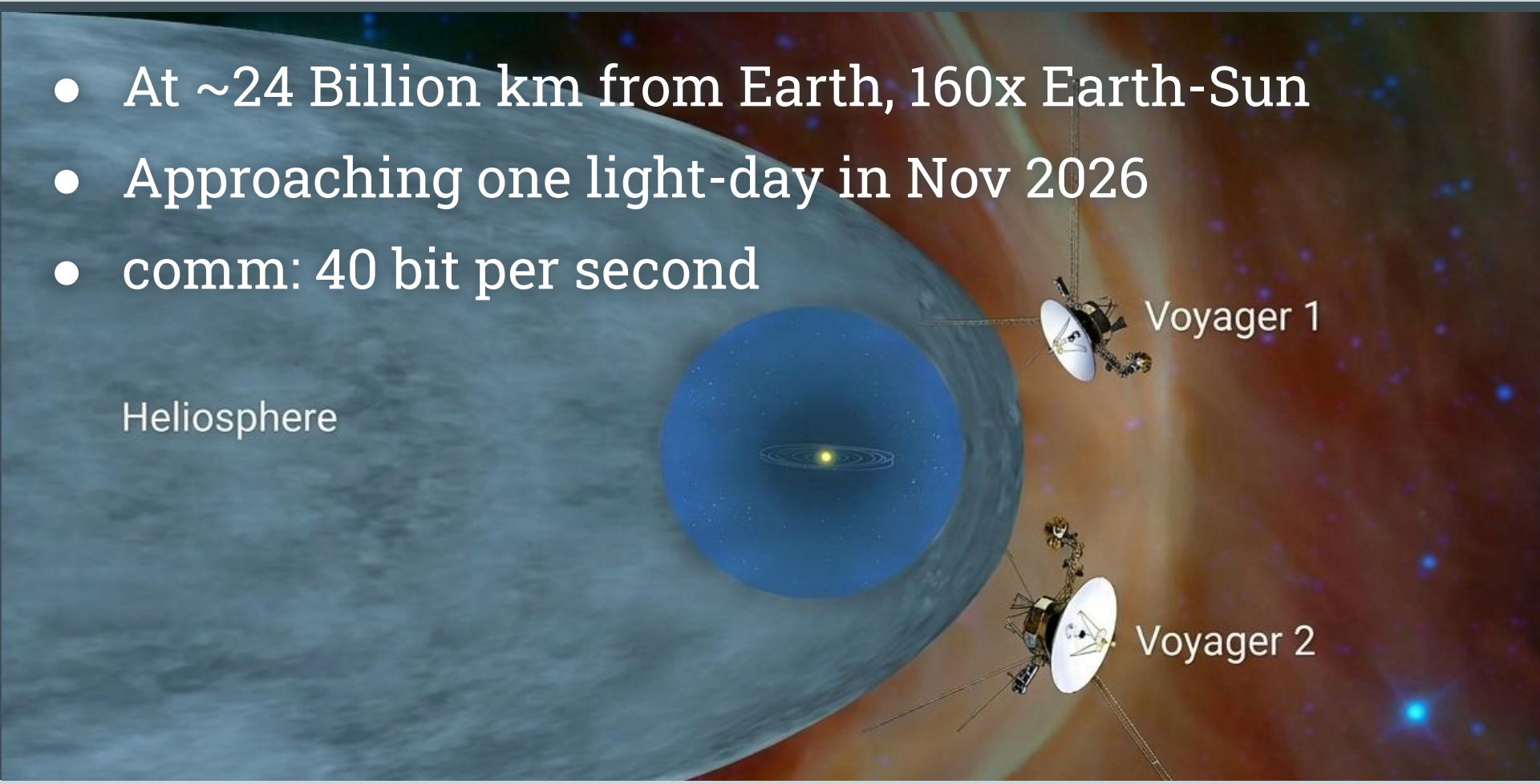


Madrid, Spain




Sun
150 million km
Earth
Moon
1.5 million km
Second Lagrange Point L2

Voyager 1974, 1976



Voyager and Deep Space Network

- Using the current distance between Earth and Voyager we get:

$$P_r = \frac{P_t G_t G_r \lambda^2}{(4\pi R)^2} = \frac{25 \times 5011.87 \times 10^7 \times (0.0357)^2}{(4\pi \times 2.37 \times 10^{13})^2}$$

- Pr=1.80×10^{-20} W, barely more than a zepto watt (zW).
- This is why large antennas are needed to receive and process the incoming signal

Starlink

- On the other hand, for Starlink, the receiving power is likely in the order of several nanowatt (nW)
- Exact numbers are difficult to calculate given the proprietary nature of Starlink

Starlink

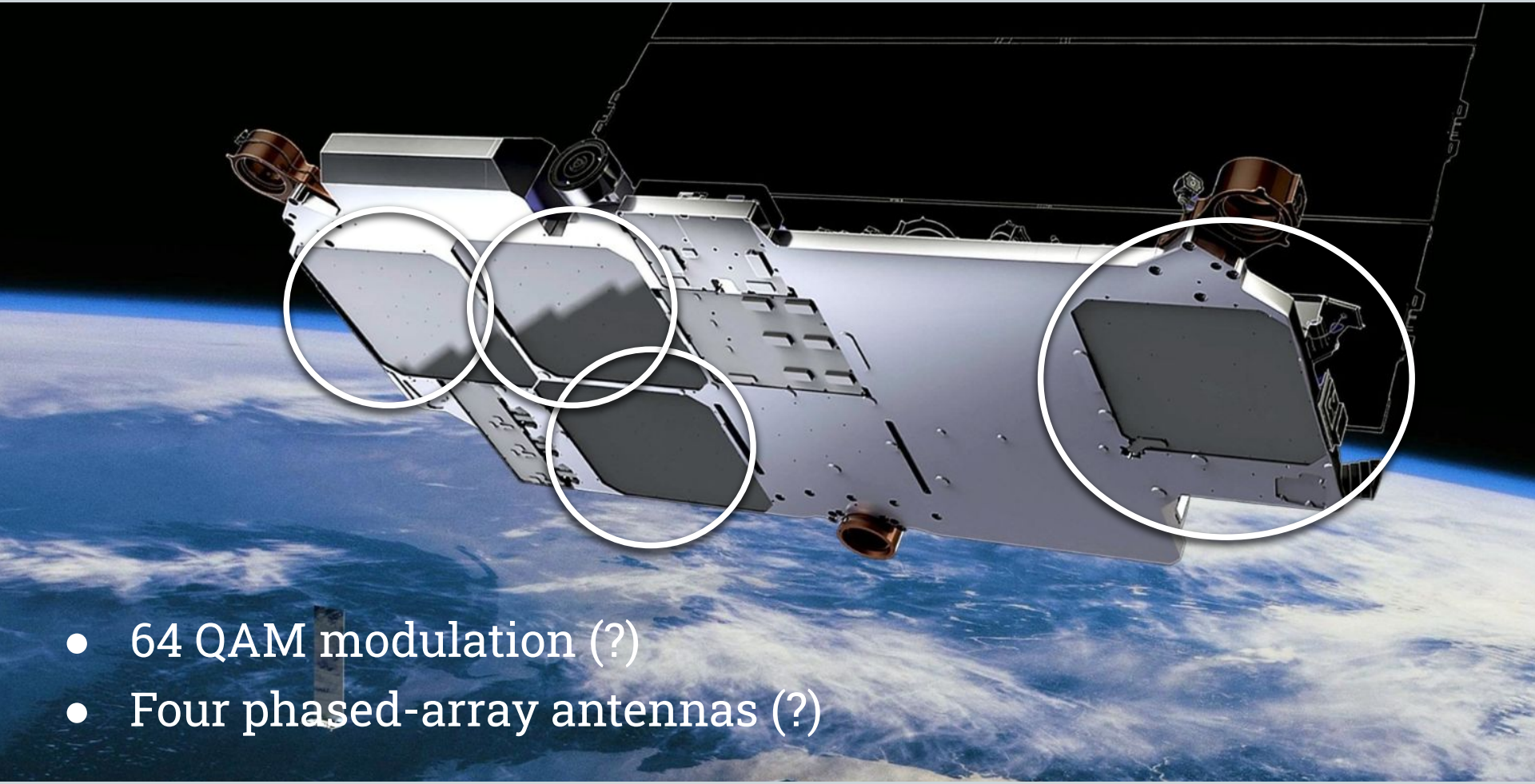
- >11,000 active satellites, ~65% of all active satellites
- LEO satellites around 550 km, also 450–490 km
  - Lifetime ~5 years or longer, followed by controlled deorbiting
  - Atmospheric drag: Requires periodic orbit corrections
- Electric propulsion for orbit maintenance
  - Solar-generated electricity to accelerate ionized gas (e.g., argon)
  - Thrust with very low propellant consumption
  - Collision avoidance, and controlled deorbiting at end of life

Starlink Capacity and Communication Technology

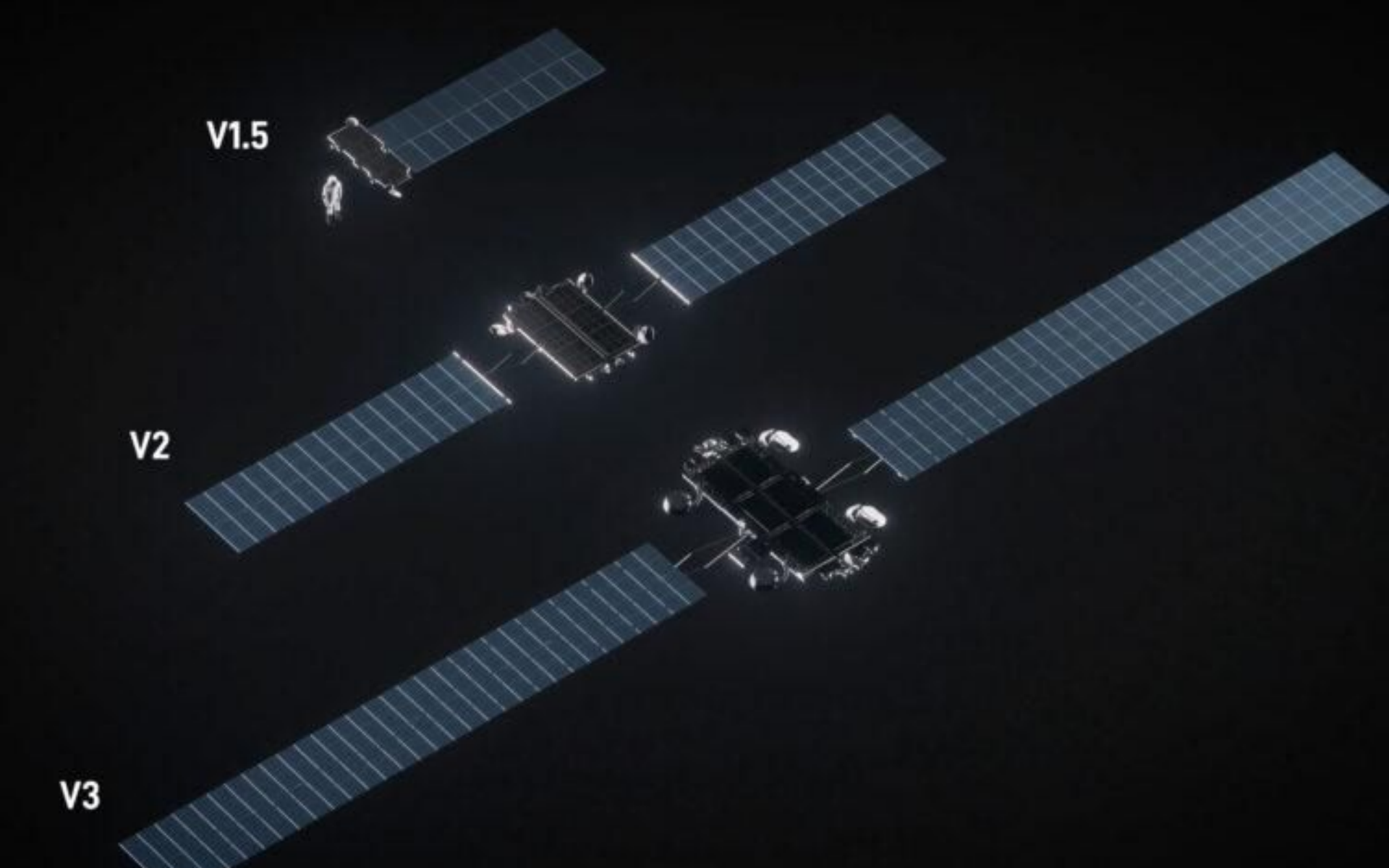
- V1.5 satellites: ~24 Gb/s downlink per satellite
- V2 Mini satellites: ~100 Gb/s per satellite
- Total constellation: > 700 Tb/s cumulative (2026)
- V3 satellites: ~1 Tb/s downlink per satellite
- Optical inter-satellite links form mesh network
  - Less reliance on ground gateways, enabling flexible routing
- Phased-array antennas
  - simultaneous beams and spatial frequency reuse

References: Technology, Starlink Updates (2026)

Starlink V1.5 satellite rendering



- 64 QAM modulation (?)
- Four phased-array antennas (?)


V1.5
V2
V3

Starlink Components

Starlink Satellites



Starlink Ground Stations



Starlink User Terminal



Image Sources: SpaceX - https://www.spacex.com/

Starlink Coverage

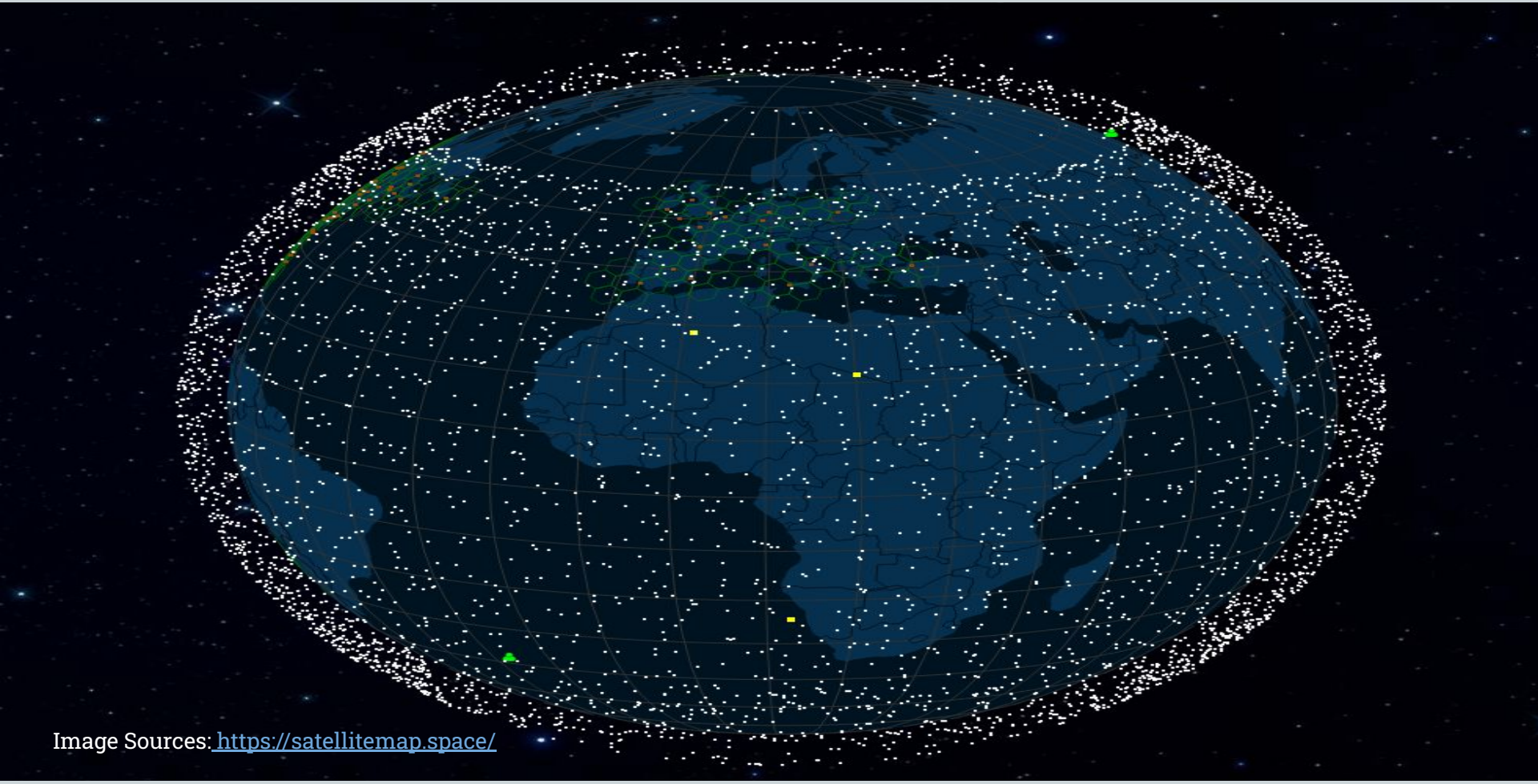


Image Sources: https://satellitemap.space/

Exercise: Global Navigation Satellite System

- Open in your course folder: 02_Notebooks/03_Mobile_Systems/ 03.3_GNSS.ipynb
  - Run the cells and answer the questions.
  - Save the completed notebook and upload it to Moodle.

